Congruence Carriers and a Sextic Reduction for Diophantine Consistency

Thirteen positive unknowns in degree 28
Fourteen positive unknowns in degree 24
Thirty-four unrestricted unknowns in degree 6

Research contribution prepared for Vladimir Reshetnikov
and the ProveIt repository

23 October 2026

Abstract

We give three fixed integer polynomials whose lack of solutions, in their specified domains, is equivalent in ordinary ZFC mathematics to the nonexistence of a finite syntactic ZFC refutation. The first has thirteen strictly positive unknowns, exact degree 28, and a literal circuit of 98 binary arithmetic operations. Its comparison point in the inspected repository has degree 34 and 93 operations at the same number of unknowns. Retaining one product as a fourteenth positive coordinate gives exact degree 24 in 101 operations, compared with the inspected fourteen-variable degree-26, 92-operation endpoint. The third is a sextic in thirty-four unrestricted integers, the sum of squares of nineteen cubic-or-lower residuals; the inspected sextic has thirty-five unknowns and twenty residuals.

For the positive construction, an existing monic Pell equation carries the fixed congruence needed to make a source gap an independent positive coordinate. This changes the order in which source bounds, divisibility, and signs become available. For the sextic, two polynomial root changes and a row reduction modulo a retained square equation allow seven registers to be eliminated from the source quartic while keeping every remaining residual cubic. We prove the new reductions and reproduce the signed auxiliary-index and squared-modulus completeness arguments.

The archive specifies coherent fixed coefficient tuples, literal polynomials, exact checkers, and source provenance. The consistency corollaries use the named inherited native compiler and refutation calibration. Finite verification establishes algebra and data integrity; the exact object-language `BProv` obligation recorded by ProveIt remains a separate formalization task. These are measured improvements over a pinned comparison snapshot, with no global minimality or global novelty claim.

Contents

1 The target and the three completed endpoints **3**
1.1 A fixed syntactic consistency statement . . . 3
1.2 What is being optimized . . . 4

2 Two general algebraic mechanisms **5**
2.1 A fixed congruence inside a monic root . . . 5
2.2 Reduce a row modulo equations which are retained . . . 6
2.3 Coordinate changes and the choice of registers . . . 6

3 Fixed coefficients and the native source interfaces **7**
3.1 Two coherent tuples . . . 7
3.2 Finite recipes, rather than an unspecified program parameter . . . 7
3.3 What the coefficient adapter supplies . . . 8

4 Signed auxiliary-index recovery **8**
4.1 Pell notation and the elementary dependencies . . . 8
4.2 The odd quotient polynomial . . . 9
4.3 Squared-modulus completeness . . . 11

5 Positive source coordinates and a congruence inside a Pell root **11**
5.1 Literal integer polynomial . . . 12
5.2 Recovery of the native congruence . . . 13
5.3 The first carrier and normalized rank recovery . . . 13
5.4 The adaptive congruence and transport sign . . . 14
5.5 Constructive completeness . . . 14
5.6 Exact degree and finite verification . . . 15

6 Fourteen positive unknowns in degree twenty-four **15**

7 A sextic in thirty-four unrestricted integers **16**
7.1 Full polynomial specification . . . 16
7.2 Reversibility of the reduction . . . 17
7.3 Why the degree is exactly six . . . 18
7.4 Finite verification and proof boundary . . . 18

8 Inherited theorems and the exact formalization boundary **19**
8.1 The ordinary mathematical dependency chain . . . 19
8.2 The exact `OBJECT_ZFC` obligation . . . 19
8.3 What exact finite verification establishes . . . 20

9 Other architectures considered **20**
9.1 A direct proof verifier instead of a blank machine . . . 21
9.2 Universal equations and integer-domain bounds . . . 21
9.3 Square carriers and free tableau padding . . . 21
9.4 Controlled packing collisions . . . 21

10 Reproduction and review **22**
10.1 Finite checks and document build . . . 22
10.2 Suggested reading order . . . 22

1 The target and the three completed endpoints

1.1 A fixed syntactic consistency statement

Fix the effective proof calculus for ZFC used by the inherited refutation calibration. Write $\mathrm{Proof}_{\mathrm{ZFC}}(p, \ulcorner \bot \urcorner)$ for its finite proof-checking relation: p codes a syntactic derivation of contradiction from ZFC axioms. A proof code includes the finite formulas and instances of the axiom schemata required to check that derivation. Define

$$\mathrm{Ref}_{\mathrm{ZFC}} :\Longleftrightarrow \exists p \in \mathbb{N}\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \ulcorner \bot \urcorner), \qquad \mathrm{Con(ZFC)} :\Longleftrightarrow \neg\, \mathrm{Ref}_{\mathrm{ZFC}}\,. \tag{1}$$

Thus the target is syntactic consistency. It does not refer to a transitive model, correctness about all arithmetical statements, or an unbounded reflection principle. The source calibration and its proof convention are part of the specification, not free choices made after examining a putative solution; see [14, 15, 16].

A *fixed* polynomial here has integer coefficients given by finite recipes and by authenticated numeral files. The only quantified parameters are the variables in its displayed witness list. Names for subexpressions are arithmetic registers, not additional existential variables. Powers occurring in a recipe for a fixed coefficient have fixed finite exponents.

Let P_{13} be the polynomial of Section 5, bound to the entire compact source-44 tuple, and let P_{14} be its product-register lift in Section 6, with the same entire tuple. Let Q_{34} be the polynomial of Section 7, bound to the entire source-34 tuple. In particular, the subscripts count unknowns, and the domains are different.

Theorem 1.1 (The completed polynomials)**.** *The following ordinary mathematical reductions hold.*

1. *P_{13} has thirteen strictly positive unknowns, exact total degree 28, and 98 live binary arithmetic gates. Its positive zeros give normalized native-source witnesses; conversely a finite refutation, through the canonical native-source construction, gives a positive zero of P_{13}.*

2. *P_{14} has fourteen strictly positive unknowns, exact total degree 24, and 101 live binary arithmetic gates. Its zero set is the graph of the uniquely determined positive coordinate $E = XY$ over the positive zero set of P_{13}.*

3. *Q_{34} has thirty-four unrestricted integer unknowns and exact degree 6. Its nineteen residuals have degree at most three. Its complete integer zero set is in explicit polynomial bijection with the zero set of the pinned forty-one-integer source quartic.*

Consequently, using the inherited source theorems identified in Section 8,

$$\exists \mathbf{x} \in \mathbb{Z}^{13}_{>0}\ P_{13}(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}, \tag{2}$$

$$\exists \mathbf{t} \in \mathbb{Z}^{14}_{>0}\ P_{14}(\mathbf{t}) = 0 \quad \Longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}, \tag{3}$$

$$\exists \mathbf{z} \in \mathbb{Z}^{34}\ Q_{34}(\mathbf{z}) = 0 \quad \Longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}\,. \tag{4}$$

Proof. The first reduction is proved by congruence recovery, Pell-index recovery, and constructive completeness in Section 5. The elementary auxiliary lemmas used there are proved in Section 4. The product-register lift is proved in Section 6. The sextic reduction follows from the two inverse coordinate maps and the retained-row identity in Section 7. Composing with the pinned native-source and finite-refutation equivalences gives (2)–(4). □

Corollary 1.2 (Ordinary ZFC-provable consistency equivalences)**.** *With the fixed proof convention and coefficients just specified, the ordinary proofs, composed with the identified source proofs, establish*

$$\mathrm{ZFC} \vdash \left(\forall \mathbf{x} \in \mathbb{Z}_{>0}^{13}\ P_{13}(\mathbf{x}) \neq 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}), \tag{5}$$

$$\mathrm{ZFC} \vdash \left(\forall \mathbf{t} \in \mathbb{Z}_{>0}^{14}\ P_{14}(\mathbf{t}) \neq 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}), \tag{6}$$

$$\mathrm{ZFC} \vdash \left(\forall \mathbf{z} \in \mathbb{Z}^{34}\ Q_{34}(\mathbf{z}) \neq 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{7}$$

Proof. Negate the existence equivalences of Theorem 1.1 and use (1). The coordinate manipulations, divisibility arguments, integer recurrences, and finite syntactic constructions are ordinary mathematics available in ZFC. This composition uses the unbounded arithmetic source proofs, not an inference from finite numerical tests or merely from an external truth statement. □

The turnstiles in Corollary 1.2 state the ordinary mathematical provability claim. They do not assert that this archive contains an exported derivation in ProveIt's particular `SetTheory.Form` calculus. Section 8.2 records the exact distinction and the still-open formal endpoint. Neither direction of the equivalence assumes that ZFC is consistent.

1.2 What is being optimized

The comparison repository is pinned at

`9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

The following table reports the relevant inspected endpoints. It is a comparison of specific artifacts, not a classification of all possible Diophantine representations. The positive predecessor is package 61's principal thirteen-variable equation, not its fourteen-variable intermediate; the integer sextic predecessor is package 62 [8, 9].

Domain	Unknowns	Degree	Gates	Endpoint or role
$\mathbb{Z}_{>0}$	13	34	93	Package 61; positive comparison point
$\mathbb{Z}_{>0}$	**13**	**28**	**98**	This report, P_{13}
$\mathbb{Z}_{>0}$	14	26	92	Package 65; a distinct tradeoff
$\mathbb{Z}_{>0}$	**14**	**24**	**101**	This report, P_{14}
$\mathbb{Z}_{>0}$	15	24	89	Package 60; a distinct tradeoff
$\mathbb{Z}$	41	4	—	Package 64; quartic source
$\mathbb{Z}$	35	6	—	Package 62; sextic comparison point
$\mathbb{Z}$	**34**	**6**	—	This report, Q_{34}
$\mathbb{Z}$	30	8	—	Package 62; a distinct tradeoff

Table 1: Measured local improvements and nearby inspected alternatives. A dash means that no optimized gate claim is being compared. Source references: [8, 9, 10, 11, 13].

For thirteen positive unknowns the degree falls by 6/34, approximately 17.65%, and the circuit gains five gates. For degree six over $\mathbb{Z}$, one unknown and one residual are removed from the comparison endpoint; the variable reduction is 1/35, approximately 2.86%. The fourteen-coordinate lift lowers degree 26 to 24 with nine additional gates. Relative to the fifteen-positive degree-24, 89-gate equation, it removes one unknown at a cost of twelve gates. The cheaper positive alternatives and the thirty-integer octic remain distinct tradeoffs.

Every binary addition, subtraction, or multiplication is charged as one gate, including multiplication by a fixed scalar and squaring. Previously computed values may be reused. Fixed numeral creation is free in this arithmetic-circuit model. It is therefore a different measurement from bit complexity, coefficient height, expanded term count, or the length of a source-language

expression. For Q_{34} the report claims exact variables, degree, residual count, and expanded term count, without a gate optimum.

The pinned collection already contains equations with fewer unknowns at much higher degrees. These endpoints do not improve those fewest-variable bounds and do not minimize every measure simultaneously. The contribution is the completed arithmetic transformations and their measured effect on the named comparison points.

2 Two general algebraic mechanisms

2.1 A fixed congruence inside a monic root

An equation which already requires an integral root can enforce an additional fixed congruence without a separate quotient variable. The following elementary observation states the exact integrality mechanism.

Lemma 2.1 (Monic fixed-congruence carrier)**.** *Let $m > 0$ and r be fixed integers. Let $H(\mathbf{x})$ be an integer polynomial, and let*

$$F(T;\mathbf{x}) = T^d + a_{d-1}(\mathbf{x})T^{d-1} + \cdots + a_0(\mathbf{x}) \in \mathbb{Z}[\mathbf{x}, T]$$

be monic in T. Define the integer polynomial

$$G(t;\mathbf{x}) = m^d F\left(t + \frac{H(\mathbf{x}) - r}{m}; \mathbf{x}\right).$$

At every integral zero of G, one has

$$H(\mathbf{x}) \equiv r \pmod{m}, \qquad T = t + \frac{H(\mathbf{x}) - r}{m} \in \mathbb{Z}, \qquad F(T;\mathbf{x}) = 0.$$

Conversely, an integral zero of F satisfying that congruence gives an integral zero of G by $t = T - (H(\mathbf{x}) - r)/m$. For strictly positive coordinates this converse also requires $t > 0$.

Proof. Multiplication by m^d clears all denominators, so G is an integer polynomial. At a zero, the displayed rational value of T is a root of a monic polynomial with integer coefficients. If $T = a/b$ in lowest terms with $b > 0$, multiplying $F(a/b;\mathbf{x}) = 0$ by b^d shows that $b \mid a^d$; hence $b = 1$. Since $t \in \mathbb{Z}$, $(H(\mathbf{x}) - r)/m \in \mathbb{Z}$, which is the required congruence. The converse is substitution, with the indicated domain condition. □

The simplest instance is a square equation:

$$(mt + H - r)^2 = m^2 W(\mathbf{x}). \tag{8}$$

Its quotient is a rational square root of an integer and is therefore an integer. This conclusion does not require coprimality of m with $H - r$, nor factoring W.

For P_{13}, the already needed strong root is written

$$N_f = 3\beta f_0 + q - \beta - 1, \qquad f = N_f/(3\beta),$$

and the strong residual is

$$N_f^2 - (3\beta)^2\{1 + \Delta(ic^2)^2\} = 0.$$

Lemma 2.1 yields

$$q = \beta(3j + 1) + 1, \qquad J = (q - 1)/\beta = 3j + 1.$$

The supplied coordinates can consequently be F, α, C, with $q = F + \alpha + C$, rather than F, α, J with a gap whose sign still needs to be proved.

This is useful because $C > 0$ immediately gives the source packing bounds once the congruence has been recovered. The first carrier no longer needs to establish the transport quotient's sign before those bounds can be used. The canonical source has $J \equiv 1 \pmod 3$ and a strong root large enough that $f_0 = f - (J - 1)/3 > 0$, which supplies the converse domain condition. The general lemma alone would not supply that positivity or the native source theorem.

2.2 Reduce a row modulo equations which are retained

The next mechanism concerns complete systems of equations. Degree can fall when a residual is replaced by an equivalent representative modulo other residuals that remain in the system.

Lemma 2.2 (Retained-row degree reduction)**.** *Let $R_1, \dots, R_s, G_1, \dots, G_t$ be integer polynomials, and choose integer polynomials H_{ij}. Put*

$$G_i' = G_i + \sum_{j=1}^{s} H_{ij} R_j \qquad (1 \le i \le t).$$

The systems

$$R_1 = \cdots = R_s = G_1 = \cdots = G_t = 0 \quad \textit{and} \quad R_1 = \cdots = R_s = G_1' = \cdots = G_t' = 0$$

have the same zero set over $\mathbb{Z}$, indeed over every commutative ring. If all retained R_j and replaced G_i' have degree at most d, their sum of squares is an integer polynomial of degree at most $2d$ whose integer zeros are exactly that common zero set.

Proof. On the locus $R_1 = \cdots = R_s = 0$, each G_i' equals G_i. Equivalently, the change of rows is a triangular polynomial matrix with diagonal entries one and inverse obtained by replacing H_{ij} with $-H_{ij}$. The sum-of-squares assertion follows because a sum of integer squares is zero exactly when all its summands are zero. □

Only the last sum-of-squares assertion uses the ordered integer domain; the row-equivalence assertion does not. The transformed sum of squares need not equal the original polynomial. It has the same integer zero set, which is the property required here.

A concrete saving is

$$\begin{aligned} R &= F_2 - f^2, \\ G &= (fo - c)^2 + g - y^2, \\ G' &= F_2 o^2 - 2foc + c^2 + g - y^2, \qquad & G' - G = o^2 R. \end{aligned} \tag{9}$$

The original G is quartic in the supplied variables; G' is cubic. The quadratic definition $R = 0$ is therefore valuable to retain, even while other definitions are eliminated.

2.3 Coordinate changes and the choice of registers

A defined register $z = E(\mathbf{x})$ can be deleted by substituting $E(\mathbf{x})$, with inverse restoration $z = E(\mathbf{x})$. This gives a bijection of zero sets, but the substituted degree may be larger. A polynomial shift of one root,

$$t_{\mathrm{old}} = t + B(\mathbf{x}),$$

is also reversible over $\mathbb{Z}$ when B does not involve t. Its purpose here is to cancel a known leading part of a norm before deleting a register.

For the integer sextic, the two useful identities are

$$\begin{aligned} (t + B)^2 - (k + B)B - 1 &= t^2 + (2t - k)B - 1, \\ (D + ac)^2 - (a^2 + 4a + 3)c^2 - 1 &= D^2 + 2acD - (4a + 3)c^2 - 1. \end{aligned}$$

The second permits deletion of the discriminant register after changing the main root. Retaining F_2, and using (9), then avoids a quartic auxiliary row. Section 7 supplies every restored coordinate and every retained equation.

The converse degree tradeoff is also useful. In P_{13}, the abbreviation $E = XY$ occurs inside expressions which are subsequently squared. Supplying E as one more positive coordinate and

retaining the equation $E - XY = 0$ lowers those degrees. Each positive old zero has exactly one new coordinate $E = XY > 0$; conversely the retained equation restores the old expression. This graph construction produces P_{14} without an additional semantic assumption, as proved in Section 6.

Monicity, polynomial coordinate changes, and elementary row operations are established algebraic tools. The specific contribution here is their placement in these arithmetic systems, including the domain and completeness proofs which make the resulting variable and degree counts valid. The report does not claim that these general algebraic facts were previously unknown.

3 Fixed coefficients and the native source interfaces

3.1 Two coherent tuples

All three endpoints use four named coefficient ports, (β, K_0, M_C, M_F), but they use different entire tuples. The compact positive tuple is source 44's complement-packed construction [5]; the unrestricted-integer tuple is source 34's mixed-radix, parity-separated construction [4]. The binding is summarized below.

	P_{13}, P_{14}	Q_{34}	Meaning
Native digit width	67	121	$\rho = 2^{\text{width}}$
Native cell length	8,807	8,801	Number of radix-ρ positions
Binary cell width d	590,069	1,064,921	$B = 2^d$, $\beta = B - 1$
Permitted native positions	986	989	Selectors, copies, anchors, dummies
Four raw numeral files	295,388 bytes	532,479 bytes	Before gzip packaging

Table 2: Complete coefficient bindings. These fixed data are inherited from the exact recipes and payloads in [5, 4].

In both recipes, if D_C, D_R denote the native current and shifted transport coefficients and $M_{F,\text{native}}$ the native high mask, the polynomial ports are

$$\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F = M_{F,\text{native}} + B - 1. \tag{10}$$

Thus the delivered M_F is already shifted. Replacing it by the unshifted mask, shifting it again, or taking individual ports from the other tuple changes the polynomial and invalidates the stated binding. The binding in `positive14/ACTIVE_BINDING.json` selects the same four numeral payloads as P_{13}; the added product register changes no coefficient.

3.2 Finite recipes, rather than an unspecified program parameter

Both recipes start from the same ordered list of 943 Wang tiles. The list is fixed; it is not generated by waiting for a machine to halt. The compiler sources also relate that literal list to the 430 literal transition actions of the chosen blank machine [4, 5, 14].

For the compact recipe, a tile with ordered labels (w, n, e, s) has payload

$$v_i = w + 2^8 n + 2^{17} e + 2^{25} s + 127 \cdot 2^{34}, \qquad c_i = 1 + 2^{10}(2^{41} - v_i).$$

Copy and anchor positions supply the 41 individual payload bits. The low ten-bit test forces either no selector or exactly one selector. Conditional on one selector, a complement sum forces its payload to equal the copied label and anchor vector. This is the local clause mechanism which permits the smaller digit width [5].

If E is that recipe's fixed support and c_e its specified weights, the current coefficient and low mask are

$$D_C = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \sum_{e\in E} c_e(\rho^{4393-e} + \rho^{5495-e}),$$
$$D_R = \rho^{42},$$
$$M_C = B - 1 - \sum_{\substack{e\in E\\ e\neq 1}} \rho^e - 2\rho^{1063}.$$

Its high mask uses the clause mask $2^{51} - 2$, the specified edge and anchor tests, and a finite upper padding mask. The retained source recipe lists every copy position and every padding exponent. The coefficient checkers reconstruct these fixed integers and compare their raw bytes; the displayed equation does not quantify over possible choices of E or of the weights.

The source-34 recipe instead uses parity-separated copy positions

$$e_t(j) = o_t + 2j, \qquad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960), \qquad 0 \le j < 10,$$

and radix-four payload weights $W_j = 2^{10+2j}$. The four edges are encoded by their occupied words, and four additional anchor fields are included. Its current shifts are $40, 41, 1100, 2200$, its transport coefficient is $D_R = \rho$, and its complete low and high masks are the ones in the source-34 recipe [4]. The different layout supports the ordinary source's integer normalizations; no assertion that the two layouts are interchangeable is needed.

3.3 What the coefficient adapter supplies

A coefficient recipe alone is not a proof that polynomial roots encode computations. The inherited adapters establish the native carry bounds, the local tile clauses, whole-cell alignment, edge matching, and the relation between the actual packed rank and the canonical Pell construction. Their completeness arguments use finite padding and a rank controller to arrange the required congruences [4, 5, 6].

For the positive reduction, the canonical compact source in particular supplies

$$q = B^N, \qquad J = 1 + B + \cdots + B^{N-1}, \qquad N \text{ odd}, \qquad R \equiv 3 \pmod 4.$$

Since $B = 2^{590069} \equiv -1 \pmod 3$, the odd geometric sum gives $J \equiv 1 \pmod 3$. This exact canonical congruence is what allows the monic carrier of Lemma 2.1 to replace the supplied coordinate J. The arbitrary-zero soundness proof subsequently recovers the required integral J; it does not assume the canonical representation $q = B^N$ for an arbitrary input.

For the integer construction, the new step is a polynomial bijection with a fixed source quartic. All sign and native completeness obligations of that source are inherited intact; the new proof does not impose an extra restriction on its zeros.

4 Signed auxiliary-index recovery

This section gives the coefficient-independent lemmas used by the positive construction. The arguments follow the pinned signed-index note in package 61 and its explicitly identified Pell sources [7, 6]. We include the proofs because the order of index recovery, rather than just the final norm identities, is material to the reduction.

4.1 Pell notation and the elementary dependencies

For an integer $A \ge 2$, put $\Delta = A^2 - 1$ and define $\chi_A(n), \psi_A(n)$ by

$$\chi_A(n) + \psi_A(n)\sqrt{\Delta} = (A + \sqrt{\Delta})^n \qquad (n \ge 0).$$

Equivalently each sequence satisfies

$$z(n+2) = 2Az(n+1) - z(n),$$

with $\chi_A(0) = 1, \chi_A(1) = A$ and $\psi_A(0) = 0, \psi_A(1) = 1$. All these values are integers. The addition formulas give

$$\chi_A(r+s) = \chi_A(r)\chi_A(s) + \Delta\psi_A(r)\psi_A(s),$$
$$\psi_A(r+s) = \psi_A(r)\chi_A(s) + \chi_A(r)\psi_A(s).$$

We use positive integer-coordinate Pell classification and strong divisibility,

$$\gcd(\psi_A(r), \psi_A(s)) = \psi_A(\gcd(r,s)). \tag{11}$$

Here $(A, 1)$ is the fundamental positive integer-coordinate solution of $x^2 - \Delta y^2 = 1$. The inherited Pell audit collects precisely these classical ingredients [6].

Lemma 4.1 (Signed abscissa step-down)**.** *Suppose $A \geq 2$, $0 < r < m$, and $t \geq 0$. If*

$$\chi_A(t) \equiv \chi_A(r) \pmod{\chi_A(m)},$$

then $t \equiv r$ or $t \equiv -r \pmod{2m}$.

Proof. Write $F = \chi_A(m)$. The addition formulas, together with $\chi_A(2m) = 2F^2 - 1$ and $\psi_A(2m) = 2F\psi_A(m)$, give

$$\chi_A(j + 2m) \equiv -\chi_A(j) \pmod{F},$$
$$\chi_A(2m - j) \equiv -\chi_A(j) \pmod{F}.$$

Reduce t modulo $2m$, and reflect an index above m. Its abscissa residue has the form $\epsilon\chi_A(s)$, where $0 \leq s \leq m$ and $\epsilon \in \{-1, 1\}$. The case $s = m$ has residue zero, whereas $0 < \chi_A(r) < F$.

If $s < m$ and $\epsilon = -1$, the congruence would force $F \mid \chi_A(s) + \chi_A(r)$. But $m \geq 2$, and the recurrence gives

$$0 < \chi_A(s) + \chi_A(r) \leq 2\chi_A(m-1) < \chi_A(m) = F.$$

Thus $\epsilon = 1$. Equality modulo F of two positive integers strictly below F is equality, and strict growth gives $s = r$. The index reductions change t only by a multiple of $2m$ and by sign, proving the assertion. □

4.2 The odd quotient polynomial

For $v \geq 0$, define the integer polynomial Q_v by

$$\chi_Z(2v+1) = ZQ_v(Z^2).$$

The relevant formulas are

$$Q_0(x) = 1, \qquad Q_1(x) = 4x - 3,$$
$$Q_{v+1}(x) = (4x-2)Q_v(x) - Q_{v-1}(x), \tag{12}$$
$$Q_v(0) = (-1)^v(2v+1), \tag{13}$$
$$Q_v(1 - A^2) = (-1)^v\psi_A(2v+1). \tag{14}$$

The recurrence follows from the Pell addition formulas. Substitution of $x = 0$ proves (13) by induction. Substitution of $x = 1 - A^2$, with the odd-index recurrence for ψ_A, proves (14); the initial cases are 1 and $1 - 4A^2$. These identities are polynomial identities over $\mathbb{Z}$, so they can be reduced modulo any positive integer.

Lemma 4.2 (Signed auxiliary-index interface)**.** *Let $A \geq 2$, $\Delta = A^2 - 1$, and let p, m, R, c, f, M be positive integers satisfying*

$$c = \psi_A(p), \qquad f = \chi_A(m), \qquad c \mid m, \qquad m > 2p,$$
$$M \geq 2, \qquad c \mid M, \qquad M^2 = \Delta(f^2 - 1),$$
$$0 < p < c/2, \qquad 0 < R < c/2.$$

If $V, y \in \mathbb{Z}$ satisfy

$$M^2V^2 - (M^2 - 1)y^2 = 1, \qquad V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}, \tag{15}$$

then $p = R$. The signs of V and y are unrestricted.

Proof. The norm equation excludes $V = 0$ and $y = 0$. Apply positive Pell classification at parameter M to the positive coordinates $(M|V|, |y|)$. There is an index $\ell \geq 1$ with

$$M|V| = \chi_M(\ell), \qquad |y| = \psi_M(\ell).$$

For even indices, the recurrence gives $\chi_M(2v) \equiv (-1)^v \pmod{M}$, which is not zero since $M \geq 2$. Thus $\ell = 2v + 1$ is odd and $|V| = Q_v(M^2)$.

As $M^2 \equiv -\Delta = 1 - A^2 \pmod{f}$, equations (14) and (15) imply

$$\psi_A(\ell) \equiv c \quad \text{or} \quad -c \pmod{f}.$$

Squaring and using $\chi_A(2j) = 1 + 2\Delta\psi_A(j)^2$, we obtain

$$\chi_A(2\ell) \equiv \chi_A(2p) \pmod{\chi_A(m)}.$$

Lemma 4.1 applies because $0 < 2p < m$. It gives $2\ell \equiv \pm 2p \pmod{2m}$, hence $\ell \equiv \pm p \pmod{m}$, and therefore modulo c.

On the other hand $c \mid M$, so $M^2 \equiv 0 \pmod{c}$. Equations (13) and (15) give $\ell \equiv \pm R \pmod{c}$. Combining the two conclusions yields $p \equiv R$ or $p \equiv -R \pmod{c}$. In the first case $|p - R| < c$, so $p = R$. The second case would make $p + R$ a positive multiple of c, although $p + R < c$. This is impossible. □

The lemma uses only its displayed hypotheses. It does not require positivity of V, a previously recovered transport quotient, or the main-root congruence used later in the positive construction.

Lemma 4.3 (Normalized strong norm supplies the divisibility)**.** *Suppose $A \geq 2$, $c = \psi_A(p) > 2$, and $f, I > 0$ satisfy*

$$f^2 - (A^2 - 1)I^2 = 1, \qquad c^2 \mid I.$$

Then for some $m \geq 1$,

$$f = \chi_A(m), \quad I = \psi_A(m), \quad pc \mid m.$$

In particular $c \mid m$ and $m > 2p$. Setting $M = (A^2 - 1)I$ gives $c \mid M$ and $M^2 = (A^2 - 1)(f^2 - 1)$.

Proof. Pell classification supplies m. By (11) and $c \mid I$, strict growth gives $p \mid m$; write $m = pv$. With $D = \chi_A(p)$, composition gives

$$I = c\psi_D(v).$$

Since $D^2 \equiv 1 \pmod{c}$, induction in the recurrence yields $\psi_D(v) \equiv vD^{v-1} \pmod{c}$ for $v \geq 1$. The integer D is invertible modulo c, and $c^2 \mid I$; therefore $c \mid v$. This proves $pc \mid m$, and $m \geq pc > 2p$. The claims about M follow directly from the norm. □

This is the normalized argument used for P_{13}, where $I = ic^2$. An older lemma for a different ordinary strong norm has an additional large-c hypothesis; that hypothesis is not needed or assumed here [7, 6].

4.3 Squared-modulus completeness

Lemma 4.4 (Explicit positive auxiliary reconstruction)**.** *Let $A \geq 2$, $\Delta = A^2 - 1$, and $R \geq 3$ with $R \equiv 3 \pmod 4$. Suppose*

$$c = \psi_A(R), \qquad f, I > 0, \qquad f^2 - \Delta I^2 = 1, \qquad c \mid I.$$

Put $M = \Delta I$. There exist $T, u > 0$ such that, for $V = f^2(cT - R) - c$,

$$M^2V^2 - (M^2 - 1)(V + u)^2 = 1.$$

Proof. Write $R = 2v + 1$. Then v is odd. Define

$$V_* = \frac{\chi_M(R)}{M} = Q_v(M^2), \qquad y_* = \psi_M(R).$$

These are positive integers, and the Pell identity gives

$$M^2V_*^2 - (M^2 - 1)y_*^2 = 1.$$

Now $M^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f^2}$. Since v is odd, (14) gives

$$V_* \equiv -c \pmod{f^2}.$$

Also $c \mid M$, so (13) gives $V_* \equiv -R \pmod c$. The strong norm gives $f^2 \equiv 1 \pmod c$, hence $\gcd(c, f^2) = 1$. It follows that

$$T = \frac{V_* + c + Rf^2}{cf^2}$$

is a positive integer: its numerator is divisible both by c and by f^2. Its definition gives $f^2(cT - R) - c = V_*$.

As $R \geq 3$ and $M \geq 2$, $V_* \geq 4M^2 - 3 > 1$. Rearranging the norm gives

$$y_*^2 - V_*^2 = \frac{V_*^2 - 1}{M^2 - 1} > 0.$$

Thus $u = y_* - V_*$ is a positive integer and $V_* + u = y_*$, which completes the construction. □

The modulus f^2 is the reason that the source quotient may be replaced by fT in the new positive equation. Completeness is proved by constructing a fresh auxiliary solution with this congruence. No divisibility of an arbitrary pre-existing auxiliary quotient is assumed.

5 Positive source coordinates and a congruence inside a Pell root

The new equation has thirteen strictly positive unknowns, exact total degree 28, and 98 live binary arithmetic operations. The degree-34, 93-operation equation of package 61 is the comparison point. The change is an existential coordinate transformation, not a same-coordinate identity. The inherited native compiler and Pell-index theorems remain explicitly identified dependencies.

The central choice is to supply the positive gap C instead of the native index J. Set $q = F + \alpha + C$; the source bounds now hold as soon as the fixed congruence on q is established. That congruence is placed inside the normalized strong Pell root. A monic square equation recovers the integrality of that root, and therefore the congruence. This permits the transport quotient's sign to be recovered at the end, removing the predecessor's degree-two positivity multiplier.

5.1 Literal integer polynomial

The positive variables are

$$F, \alpha, C, f_0, i, T, X, Y, r, \eta, \zeta, u, \gamma.$$

Use the whole compact source-44 tuple (β, K_0, M_C, M_F), with $\beta = 2^{590069} - 1$ and the already shifted high mask M_F. Put

$$B = \beta + 1, \qquad b = 3\beta, \qquad L = \beta b^2.$$

These are fixed numerals. The following names are polynomial abbreviations, not additional witnesses:

$$\begin{aligned}
q &= F + \alpha + C, & k &= \eta + \zeta, & c &= kY + \eta,\\
E &= XY, & \ell &= EkY, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & I &= ic^2,\\
N_f &= bf_0 + q - B, & F_2 &= N_f^2, & M_0 &= q^3(q-1),\\
S_\beta &= 4\beta qY, & M_\beta &= qS_\beta.
\end{aligned}$$

The remaining registers are

$$\begin{aligned}
G &= (q-1)(q-F) + \alpha,\\
\widehat{R} &= \beta G(q^2-1) + (M_C + qM_F)(q-1),\\
B_t &= (qK_0 + X)C - qF,\\
Z &= M_0(\ell + r) + (q-1)Y - q^3 B_t,\\
\widehat{D} &= M_\beta(X + ac) + H\{M_\beta\gamma + S_\beta X - \beta(k-1) + \widehat{R}\},\\
\widehat{V} &= F_2(\beta cT - \widehat{R}) - Lc,\\
\widehat{Y} &= \widehat{V} + Lu.
\end{aligned}$$

Define four integer residuals

$$A_0 = Z^2 - M_0^2\{\ell(\ell + k) + 1\}, \tag{16}$$

$$A_1 = \widehat{D}^{\,2} - M_\beta^2(1 + \Delta c^2), \tag{17}$$

$$A_a = b^2\widehat{Y}^{\,2} - \Delta(F_2 - b^2)(\widehat{Y}^{\,2} - \widehat{V}^{\,2}) - b^2L^2, \tag{18}$$

$$A_s = F_2 - b^2(1 + \Delta I^2). \tag{19}$$

The equation is

$$\boxed{P_{13} = A_0^2 + A_1^2 + A_a^2 + A_s^2 = 0.} \tag{20}$$

Every coefficient in this definition is an integer. There are no variable exponents or unspecified program parameters.

For the proof only, use the rational abbreviations

$$\begin{aligned}
f &= N_f/b, & J &= (q-1)/\beta, & R &= \widehat{R}/\beta,\\
S &= 4qY, & M &= qS = 4q^2Y, & v &= k - R - 1,\\
D_* &= M(X + ac) + H(M\gamma + SX - v),\\
V &= f^2(cT - R) - c, & U &= \Delta(f^2 - 1).
\end{aligned}$$

Then $\widehat{D} = \beta D_*$, $\widehat{V} = LV$, and the residuals have the exact interpretations

$$\begin{aligned}
A_s &= b^2(f^2 - \Delta I^2 - 1),\\
A_1 &= \beta^2\{D_*^2 - M^2(1 + \Delta c^2)\},\\
A_a &= b^2L^2\{(V+u)^2 - Uu(2V+u) - 1\}.
\end{aligned}$$

These identities clear fixed denominators; they do not introduce divisions into the delivered polynomial.

5.2 Recovery of the native congruence

Suppose (20) holds. All four residuals vanish separately. As $q \geq 3$ and $f_0 \geq 1$,

$$N_f \geq 2\beta + 2 > 0.$$

Equation (19) makes $(N_f/b)^2 = 1 + \Delta I^2$ an integer. A rational square root of an integer is integral. Hence $f = N_f/b$ is a positive integer and

$$q - \beta - 1 = 3\beta j, \qquad J = 3j + 1$$

for an integer j. The inequality $q > 0$ excludes $j < 0$. Thus $J \geq 1$, $q \geq B \geq 16$, and, since $\beta \equiv 1 \pmod 3$, $q \equiv 2 \pmod 3$. Also R is now integral with its exact native formula.

All of F, α, C are positive and $q = F + \alpha + C$. The authentic mask bounds give

$$3q < R < q^4 - 2. \tag{21}$$

For example, $G = q\alpha + (q-1)C$ lies between q and $q^2 - q$, while $0 < (M_C + qM_F)J < (q-1)(1+2q)$. These estimates use no transport quotient.

5.3 The first carrier and normalized rank recovery

Equation (16) makes $\tau = Z/M_0$ an integer square root of $\ell(\ell + k) + 1$, possibly with negative sign. Reducing $M_0 \mid Z$ modulo q^3 and $q - 1$ gives

$$q^3 \mid Y, \qquad q - 1 \mid B_t.$$

Write $s = Y/q^3 > 0$ and $t = B_t/(q-1) \in \mathbb{Z}$. Thus

$$\tau = \ell + r + s - t, \qquad \tau^2 - \ell(\ell + k) = 1.$$

Neither t nor τ is assumed positive at this stage.

Similarly, (17) gives an integer $g = D_*/M$ with $g^2 - \Delta c^2 = 1$. It is positive: $v < k$, $c > kY$, $H < 5a$, and $MY > 5$ imply

$$D_* > Mac - Hk > ak(MY - 5) > 0.$$

Put $A = a + 2$ and $P_0 = 2XY^2 + 1$. Pell classification supplies $n, p \geq 1$:

$$|\tau| = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n), \qquad g = \chi_A(p), \quad c = \psi_A(p).$$

The usual recurrence gives $P_0 > A$; since $c > k$, monotonicity implies $n < p$. If $n = 1$, then $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$, whereas $p \geq 2$ forces $c \geq 2A \geq 4Y + 4$. Hence $n \geq 2$, and

$$c > kY \geq 4P_0Y > 8XY^3 \geq 8q^9 > 2R, \qquad c > 2p.$$

The *normalized* strong equation is essential here. It gives $f = \chi_A(m)$ and $I = \psi_A(m)$. Since $I = ic^2$, strong divisibility gives $p \mid m$. Write $m = pv_0$. Composition and reduction modulo c give

$$I/c = \psi_g(v_0) \equiv v_0 g^{v_0 - 1} \pmod c.$$

Because $g^2 \equiv 1 \pmod c$, it follows that $c \mid v_0$, hence $c \mid m$ and $m \geq pc > 2p$. This argument does not require the large-c hypothesis of the separate ordinary-strong rank lemma.

Set $M_a = \Delta I$. Then $M_a^2 = U$, $c \mid M_a$, and (18) becomes

$$M_a^2 V^2 - (M_a^2 - 1)(V + u)^2 = 1.$$

Both displayed Pell coordinates are nonzero. The definition of V and $f^2 \equiv 1 \pmod{c}$ give

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}.$$

All hypotheses of the inherited signed auxiliary-index theorem are now established, including $0 < p, R < c/2$ and $c \mid m$. That theorem yields $p = R$. No deleted quotient or transport sign has been used.

5.4 The adaptive congruence and transport sign

The main carrier gives $M \mid H(SX - v)$. Since $q \mid Y$ and $q \equiv 2 \pmod 3$, we have $\gcd(q, H) = 1$. Moreover, H is odd and $\gcd(Y, H) \mid 3$, so

$$\mu = \frac{S}{\gcd(S, H)} \geq \frac{4qY}{3} > q^4 > R + 1.$$

Reduction modulo S gives $\mu \mid v$. Both S and M divide $4XY^2$ already, because $q^3 \mid Y$; no assumption $q \mid X$ is needed. The first Pell recurrence gives $k \equiv 2n \pmod{4XY^2}$. Thus $\mu \mid 2n - R - 1$. As $n < p = R$ and $|2n - R - 1| < R < \mu$,

$$2n = R + 1.$$

Therefore $v = k - 2n$ is a positive multiple of $4XY^2$, in particular of M. The main congruence now gives $q \mid HX$, hence $q \mid X$. The integers

$$w = X/q > 0, \qquad h = v/E > 0, \qquad \gamma_{\mathrm{old}} = \gamma + w - v/M$$

restore the source scale and index quotient. The last one is positive: $n \geq 2$ implies $c > X$, while $g > (a + 1)c$, so $H\gamma_{\mathrm{old}} = g - ac - X > c - X > 0$.

After $X = qw$, $B_t = q((K_0 + w)C - F)$. As $q - 1 \mid B_t$ and $\gcd(q, q - 1) = 1$, the number

$$z_{\mathrm{old}} = \frac{(K_0 + w)C + q - F - 1}{q - 1}$$

is integral. Its numerator is positive, since $K_0, w, C > 0$ and $q - F = \alpha + C \geq 2$. Thus $z_{\mathrm{old}} > 0$ and the authentic transport equation is restored. This argument allows $z_{\mathrm{old}} - 1 = 0$ and does not assume $w > q$ for arbitrary solutions.

Finally $T_{\mathrm{old}} = fT$ and $y = |V + u|$ restore the remaining positive source coordinates. The inherited native compiler extracts a finite syntactic ZFC refutation.

5.5 Constructive completeness

Conversely, start with canonical positive normalized source witnesses for a refutation. Supply their F, α, C and put

$$j = (J - 1)/3, \qquad f_0 = f - j.$$

The first is integral, and $f_0 > 0$ because $f \geq a + 2 > Y \geq q^3 > q > j$. This gives $N_f = bf$ and $A_s = 0$. The canonical odd-index reconstruction from package 61 supplies fresh $T, u > 0$ with

$$V_* = \frac{\chi_{M_a}(R)}{M_a}, \quad y_* = \psi_{M_a}(R), \quad T = \frac{V_* + c + Rf^2}{cf^2}, \quad u = y_* - V_*.$$

Its integrality uses the established squared-modulus congruences and $R \equiv 3 \pmod 4$. It gives $A_a = 0$.

Let $t = q(z_{\mathrm{old}} - 1) \geq 0$ and set

$$r = \tau - \ell - s + t, \qquad \gamma = \gamma_{\mathrm{old}} + v/M - w.$$

Both are positive. For the first, $n \geq 2$ gives $k \geq 8XY^2 + 4 > 3s$, and

$$\tau^2 - (\ell + k/3)^2 = k^2(XY^2/3 - 1/9) + 1 > 0.$$

Hence $\tau - \ell > k/3 > s$. For the second, v is a positive multiple of $4XY^2$, so $v/M \geq XY/q^2 = wY/q \geq wq^2 > w$. Substitution gives $Z = M_0\tau$ and $D_* = Mg$, establishing $A_0 = A_1 = 0$. This proves the reverse existence implication. Composition with the native syntactic calibration and negation gives the claimed equivalence with Con(ZFC), by ordinary mathematics formalizable in ZFC.

5.6 Exact degree and finite verification

The cancellations

$$\begin{aligned} Z &= M_0\ell + Z_t, \\ A_0 &= 2M_0\ell Z_t + Z_t^2 - M_0^2(\ell k + 1), \\ (X + ac + H\gamma)^2 - (1 + \Delta c^2) &= 2ac(X + H\gamma) + (X + H\gamma)^2 - Hc^2 - 1, \\ \widehat{Y}^{\,2} - \widehat{V}^{\,2} &= Lu(2\widehat{V} + Lu) \end{aligned}$$

give residual degrees $(13, 13, 13, 14)$. The degree-14 homogeneous part of A_s is

$$-9\beta^2 i^2(\eta + \zeta)^4 X^2 Y^6.$$

Only A_s^2 reaches degree 28, so the leading form of P_{13} is

$$\boxed{81\beta^4 i^4(\eta + \zeta)^8 X^4 Y^{12}.}$$

The coefficient of $i^4\eta^8 X^4 Y^{12}$ is $81\beta^4 > 0$ for the actual fixed instance. The exact degree is therefore unaffected by specialization of the other three native ports.

The independent standard-library checker expands the four literal residuals into respectively 1921, 5565, 26877, and 81 symbolic monomials, compares separately specified formulas, checks their exact witness degrees, and verifies the final sum of squares over independent residual indeterminates. It also checks all wires and liveness. The counts are 25 additions, 16 subtractions, and 57 multiplications: 98 operations, or 91 for the four-residual system. The normal and optimized Python runs both pass. These finite checks certify the emitted algebra and counts; they neither decide consistency nor replace the unbounded mathematical argument and its named inherited dependencies.

6 Fourteen positive unknowns in degree twenty-four

An elementary register lift of P_{13} gives a useful third endpoint: fourteen strictly positive unknowns, exact degree 24, and 101 live binary arithmetic operations. Retain all thirteen variables from Section 5 and introduce a positive variable E. In that section's formulas, use this supplied E wherever the abbreviation $E = XY$ occurred. Thus

$$a = E + Y, \qquad \ell = E(\eta + \zeta)Y, \qquad \Delta = (E + Y)^2 + 4(E + Y) + 3.$$

Let $\widetilde{A}_0, \widetilde{A}_1, \widetilde{A}_a, \widetilde{A}_s$ be the resulting four residuals, and set

$$\boxed{P_{14} = \widetilde{A}_0^2 + \widetilde{A}_1^2 + \widetilde{A}_a^2 + \widetilde{A}_s^2 + (E - XY)^2 = 0.} \tag{22}$$

The coefficient tuple is exactly the whole compact tuple already bound to P_{13}.

Exact equivalence. Every positive zero x of P_{13} extends to the positive zero (x, XY) of P_{14}. Conversely, a positive zero of P_{14} satisfies $E = XY$, since every squared residual vanishes separately. Substitution then recovers $P_{13}(x) = 0$. The extension is unique, so these maps give a bijection of positive zero sets. This elementary argument is formalizable in ZFC and adds no mathematical source assumption to the preceding consistency reduction.

Degree and circuit size. With E independent, the five residual degrees are respectively

$$12, \quad 11, \quad 12, \quad 12, \quad 2.$$

Only $\widetilde{A}_s$ contains i, and its highest homogeneous part is

$$-9\beta^2 i^2(\eta+\zeta)^4 Y^4(E+Y)^2.$$

Consequently the coefficient of $i^4\eta^8E^4Y^8$ in P_{14} is $81\beta^4 \neq 0$. The exact total degree is therefore 24. The former multiplication XY remains live in the new defining residual. One subtraction, one square, and one addition extend the 98-operation circuit to 101 operations: 26 additions, 17 subtractions, and 58 multiplications. The delivered JSON circuit, deterministic generator, exact symbolic checker, and independent audit make this tradeoff reproducible. The register lift is elementary; the substantive new arithmetic mechanism remains the thirteen-variable construction.

7 A sextic in thirty-four unrestricted integers

The second construction improves the inspected degree-six integer equation from 35 unknowns and 20 residuals to 34 unknowns and 19 residuals. Its starting point is the 41-integer quartic of package 64 at the pinned repository revision. The new step is an explicit bijection of its complete integer zero set, so no new Pell or positivity hypothesis is required. The inherited quartic's compiler and refutation calibration remain the semantic dependencies. No global minimality or new kernel certificate is claimed.

7.1 Full polynomial specification

Use the whole source-34 coefficient tuple (β, K_0, M_C, M_F), with $\beta = 2^{1064921} - 1$ and the already shifted high mask M_F. This is a different, coherent tuple from the compact tuple used in the thirteen-positive construction; individual ports are not interchanged. The 25 arithmetic unknowns are

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j, E, c, a, D, L, F_2, g, I, u.$$

There are also three triples of unknowns (a_1, a_2, a_3), (b_1, b_2, b_3), and (e_1, e_2, e_3). All 34 unknowns range over *unrestricted integers*. Define the abbreviation

$$\Pi(x, y, t) = 1 + x^2 + y^2 + t(t+1)$$

and the polynomial registers

$$q = \beta J + 1, \qquad k = \eta + \zeta, \quad C = q - F - \alpha, \quad H = 4a + 3,$$
$$R_L = \big((q-1)(q-F) + \alpha\big)L + (M_C + qM_F)J.$$

In particular R_L uses the supplied coordinate L, not a substituted degree-two expression for it. This choice is material to the degree bound.

The first seven residuals are

$$\begin{aligned}
Q_1 &= L - q^2 + 1,\\
Q_2 &= 2u^2 - 4wq,\\
Q_3 &= a - E - (2v+1)q(L+1),\\
Q_4 &= E(2u^2+1) - 2u^2 a,\\
Q_5 &= c - \eta - (a-E)k,\\
Q_6 &= \tau^2 + (2\tau - k)E(c-\eta) - 1,\\
Q_7 &= D - 2u^2 - \gamma H.
\end{aligned}$$

The next six are

$$\begin{aligned}
Q_8 &= D^2 + 2acD - Hc^2 - 1,\\
Q_9 &= k - 1 - hE - R_L,\\
Q_{10} &= (K_0 + 4w)C - F - z(q-1),\\
Q_{11} &= F_2 - f^2,\\
Q_{12} &= fo - c - jc + k - 1 - hE,\\
Q_{13} &= g + F_2 o^2 - 2foc + c^2 - y^2.
\end{aligned}$$

Finally put

$$\begin{aligned}
Q_{14} &= I - (2r+1)c^2,\\
Q_{15} &= I^2 - (a^2 + 4a + 3)(F_2 - 1),\\
Q_{16} &= y^2 - I^2 g - 1,\\
Q_{17} &= F\alpha - \Pi(a_1, a_2, a_3),\\
Q_{18} &= FC - \Pi(b_1, b_2, b_3),\\
Q_{19} &= \eta\zeta - \Pi(e_1, e_2, e_3).
\end{aligned}$$

The single equation is

$$\boxed{Q_{34} = \sum_{t=1}^{19} Q_t^2 = 0.} \tag{23}$$

The JSON packet calls v, w respectively `v_scale`, `w_scale`; its three guard triples are the corresponding `FA_pos`, `FC_pos`, and `eta_zeta_pos` coordinates. The displayed τ, D are *new* root coordinates, as the restoration formulas below specify.

7.2 Reversibility of the reduction

Set $B_0 = E(c - \eta)$ for this proof. Given a zero of (23), restore the seven deleted source registers by

$$\begin{aligned}
P &= k + B_0, & X &= 2u^2, & N &= q(L+1),\\
C_2 &= c^2, & \Delta &= a^2 + 4a + 3, & V &= fo - c, & B_{\mathrm{aux}} &= I^2,
\end{aligned}$$

and restore the source roots by

$$\tau_{\mathrm{old}} = \tau + B_0, \qquad D_{\mathrm{old}} = D + ac.$$

The source's scale coordinate is w here, so its arithmetic occurrence of the old scale is $4w$, exactly as in package 64's quartic. Every restoration is an integer polynomial; no divisibility condition has been discarded.

The seven defining source rows vanish identically. The two root shears give the exact identities

$$(\tau + B_0)^2 - (k + B_0)B_0 - 1 = \tau^2 + (2\tau - k)B_0 - 1,$$
$$(D + ac)^2 - (a^2 + 4a + 3)c^2 - 1 = D^2 + 2acD - (4a + 3)c^2 - 1.$$

Thus the apparent degree-four terms cancel before the sum of squares is formed. The one remaining nontrivial row replacement is

$$Q_{13} - ((fo - c)^2 + g - y^2) = o^2(F_2 - f^2) = o^2 Q_{11}. \tag{24}$$

Because Q_{11} is retained, the replaced row and its source row vanish together. This is an invertible elementary row operation on the system, not division by a variable or passage to rational witnesses. Every other restored source residual is exactly its listed Q_t.

Conversely, given any integer zero of the source quartic, forget its seven defined registers, set

$$\tau = \tau_{\text{old}} - E(c - \eta), \qquad D = D_{\text{old}} - ac,$$

and keep all other coordinates. The same identities establish every listed residual. These projection and restoration maps are inverse on the complete integer zero sets. Therefore the inherited source theorem gives, by ordinary mathematics formalizable in ZFC,

$$(\exists \mathbf{x} \in \mathbb{Z}^{34} : Q_{34}(\mathbf{x}) = 0) \quad \Longleftrightarrow \quad \text{there exists a finite syntactic ZFC refutation.}$$

Negating yields the required unsolvability/consistency equivalence.

7.3 Why the degree is exactly six

All Q_t have degree at most three when the four coefficient ports are fixed. In particular retaining $F_2 = f^2$ allows the square of $fo - c$ to be represented by $F_2 o^2 - 2foc + c^2$, which is cubic. Eliminating F_2 instead would lose this saving. The simultaneous shear of the main root allows Δ to be eliminated while retaining that square register. The construction deletes seven coordinates and seven equations from the 41-coordinate, 26-row source, giving the claimed 34 and 19.

Only $Q_{16} = y^2 - I^2 g - 1$ contains both I and g. Hence

$$[I^4 g^2] Q_{34} = 1.$$

This certifies exact total degree six even after authentic specialization of all four fixed coefficients. The expansion with coefficient ports kept symbolic contains 368 terms. No optimized arithmetic-gate claim is made for this endpoint.

7.4 Finite verification and proof boundary

The self-contained standard-library programs verify every source restoration, the correction (24), all 19 independently transcribed formulas, liveness of the 34 variables, and the unit degree certificate. They also reconstruct the entire four-numeral source-34 tuple from the literal 943-tile list and compare its bytes with the retained payloads; the four raw integers occupy 532,479 bytes. Normal and optimized Python runs both pass.

These checks establish exact finite algebra and coefficient integrity. The unbounded compiler and consistency calibration are the named inherited mathematical results, and no new end-to-end proof-assistant certificate is claimed. The improvement is one unrestricted variable at degree six, rather than a change in the fewest-variable high-degree bounds.

8 Inherited theorems and the exact formalization boundary

8.1 The ordinary mathematical dependency chain

There are three distinct levels of justification in this package. First, the new algebraic reductions are stated and proved in this report. The sextic reduction is a zero-set bijection over all integers; the positive reduction additionally uses classical Pell arithmetic and the canonical source witness promises explicitly listed above. Second, the arithmetic source interfaces relate their zeros to finite native tableaux. Third, the literal machine/tableau calibration relates the permitted finite computations to finite syntactic refutations in the specified ZFC calculus [4, 5, 10, 14].

The inherited components used in the consistency corollaries are:

1. **The complete coefficient adapters.** The recipes include carry bounds, exact native supports, local clauses, alignment, matching conditions, and canonical rank control. A numerical inequality for each mask does not substitute for this whole interface.

2. **The Pell/native arithmetic kernel.** Soundness decodes any source zero, while completeness constructs suitable positive or signed source coordinates from a finite accepted tableau. The audit in [6] identifies which Pell lemmas are used and how normalized and ordinary strong equations differ.

3. **The syntactic refutation calibration.** The fixed blank computation is calibrated to the actual finite proof relation. This is the semantic link needed for $\mathrm{Ref}_{\mathrm{ZFC}}$, and is preserved by the new coordinate changes.

The pinned native-calibration note also separates two formal milestones [15]. It records the external theorem `KEndpoint215.blank_halts_iff_zfc_refutation`, relating the literal blank machine to `BProv ZFCax_s [] Form.fBot`. The formal bridge from the particular coherent native coefficient instance to that blank-halting predicate remains a distinct contract. The cited native and Pell sources give ordinary mathematical proofs; this report does not claim to complete that native Lean contract.

This is a composition of ordinary mathematical proofs. The report does not reprove every transitive source theorem, and the exact finite checkers do not discharge those unbounded dependencies. The source manifest identifies the retained files and their pins so that this boundary is inspectable. There is no requirement to choose a coefficient by deciding consistency or by running an unbounded search.

8.2 The exact `OBJECT_ZFC` obligation

ProveIt's pinned `OBJECT_ZFC.md` makes a precise distinction between an external Lean theorem and an internal derivation in its chosen ZFC calculus [16]. In that development, the external theorem `ExplicitZFC.noSyntaxZero_iff_conZFC` concerns the actual external predicate

```
BProv ZFCax_s [] Form.fBot.
```

It is not already an object-language derivation of a sentence expressing the same equivalence.

The file records the still-open exact target, for that development's fixed nine-variable polynomial:

```
BProv ZFCax_s []
  (fIff ObjectCodeSeal.consistencySentence
        ObjectEquation.unsolvabilitySentence).
```

Both sides have been supplied as actual `SetTheory.Form` sentences, with arithmetic interpretation infrastructure. The missing endpoint is the displayed derivation. Our P_{13}, P_{14}, and Q_{34} are different supplied polynomials, so a corresponding certified endpoint for any of them would also have to instantiate its own equation sentence and prove the required internal interpretation.

The obstacle cannot be removed by checking standard numerals alone. An internal ZFC model has its own natural numbers and its own proof codes, including nonstandard ones. An external theorem quantified over the ambient natural numbers does not automatically give correctness for all those internal objects. The pinned file identifies the needed route: establish the internal refutation-checking interpretation and the arithmetic representation in arbitrary models, or produce object-language derivations directly, then apply the available completeness machinery [16].

Corollary 1.2 is the ordinary mathematical claim obtained by the stated source composition. This archive supplies no new end-to-end Lean/Rocq certificate and no exported `BProv ZFCax_s []` proof of that claim. The new algebra would be a small and explicit part of such a future formalization; the inherited compiler and internal proof-coding interpretation remain the larger obligations.

8.3 What exact finite verification establishes

For P_{13}, the delivered circuit and independent symbolic interpretations establish the four exact residual identities, their degrees, the fixed-slice leading coefficient, all live inputs and gates, and the final sum of squares. The four residual degrees are

$$(13, 13, 13, 14),$$

and their symbolic-port monomial counts are

$$(1921, 5565, 26877, 81).$$

The 98 operations consist of 25 additions, 16 subtractions, and 57 multiplications. The leading form

$$81\beta^4 i^4(\eta + \zeta)^8 X^4 Y^{12}$$

shows exact degree 28 on the authentic fixed slice, not just a formal degree upper bound before coefficient specialization.

For P_{14}, the five residual degrees are

$$(12, 11, 12, 12, 2).$$

The unique extension $E = XY$ is checked against the thirteen-coordinate packet. The coefficient of $i^4\eta^8 E^4 Y^8$ is $81\beta^4 > 0$, which proves exact degree 24. Its 101 gates comprise 26 additions, 17 subtractions, and 58 multiplications; the five-residual system uses 92 gates before its finalizer.

For Q_{34}, exact expansion checks all nineteen independently specified residuals, the seven restored registers, both root changes, and the retained-row identity. The coefficient $[I^4 g^2]Q_{34} = 1$ certifies exact degree six independently of the four fixed ports. Its symbolic-port expansion has 368 terms. Complete tuple reconstruction and raw-byte comparisons establish the finite coefficient binding.

The programs are ordinary executable algebra checks. Their successful runs do not prove that the polynomial has no zero, exhaust its infinite domain, or settle Con(ZFC). They verify the finite formulas and data to which the unbounded mathematical arguments apply. The independent review notes are internal audits with their scope stated; they are not external peer review.

9 Other architectures considered

The construction problem does not require a particular Turing machine or Jones' universal expression. We considered routes which could change the representation more fundamentally. None produced a stronger completed bound within this work. The following distinctions explain why the proved reductions were retained as the deliverable.

9.1 A direct proof verifier instead of a blank machine

A finite ZFC proof checker can be treated as a syntactic computation without first selecting K215. One could encode formula parsing, axiom-instance checking, substitution, and inference directly into local constraints, or first pass through a small rewriting or combinatory system. Such a construction would target the desired finite-refutation predicate from the beginning.

The size of a verifier description, however, does not directly determine the number or degree of its final Diophantine unknowns. A complete improvement would need an explicit bounded local alphabet, a uniform packing theorem, sound handling of arbitrarily long proofs, and the actual fixed coefficients. Replacing a machine constant while keeping the same arithmetic skeleton can change coefficient height without changing the witness count or degree at all. We did not obtain a smaller fully calibrated polynomial by this route. It remains useful for the internal proof-coding task described in Section 8.2.

9.2 Universal equations and integer-domain bounds

Universal Diophantine compilation is a classical baseline [1]. Recent work by Bayer, David, Hassler, Matiyasevich, and Schleicher studies explicit universal bounds over the integers alongside formalization [2]; Sun's pinned September 2026 preprint proves undecidability with ten unrestricted integer unknowns [3]. These are relevant to the high-degree, few-variable end of the subject.

The endpoints proved here do not use a new Jones or Sun reduction. Their arithmetic skeletons still inherit the fixed native/refutation calibration. An application of a universal theorem alone would not establish that its fixed consistency instance improves the degree, coefficient height, or literal circuit cost of the measured endpoints. The literature references provide context, not a novelty certificate for the transformations in this report.

9.3 Square carriers and free tableau padding

There is a tempting alternative for integer positivity guards. Every residue $a \equiv 1 \pmod 8$ has an odd square root modulo 2^k for $k \geq 3$, and a chosen root gives arbitrarily large squares in that residue class. The elementary lift follows because, for odd y,

$$(y + 2^{j-1})^2 - y^2 \equiv 2^j \pmod{2^{j+1}} \qquad (j \geq 3).$$

Thus, if a computation encoding ignored a sufficiently flexible high prefix, one might try to move a positivity requirement onto a square without an additional witness triple.

The existing native word is constrained throughout its masked support. The source gap C is a bounded complete word, not merely a low residue with arbitrary unused high bits. Arbitrary high digits in a square can violate local masks, copy fields, and transport relations. The harmless dummy coordinates used for canonical rank adjustment do not by themselves prove the freedom needed by this square argument. No positivity guard was removed on that unsupported premise.

9.4 Controlled packing collisions

Lemma 2.2 suggests a further direction: a packed coefficient need only recover a sufficient generating family of residual equations, rather than isolate every residual literally. If a collision contributes a polynomial multiple of an equation that is retained elsewhere, it may preserve the zero set. This is a concrete algebraic possibility, as (9) demonstrates in the sextic.

Extending the idea to a few-variable packing requires additional proofs about coefficient extraction, signs, and carries at the actual digit base. Row-equivalence over a polynomial ring alone does not authorize a collision in an integer digit encoding. No improved nine-variable equation is claimed from that prospective extension. The sextic gives a completed instance of the retained-equation principle with all inverse maps explicit.

10 Reproduction and review

10.1 Finite checks and document build

From the unpacked archive root, run

```
python3 verify_all.py
python3 -O verify_all.py
```

The supplied checkers use the Python standard library. Source identity, coefficient reconstruction, literal formula interpretation, exact degree certificates, and zero-set restoration checks are separate parts of the replay. The optimized invocation checks that essential validation is not silently removed with Python assertions. The consolidated receipts are `VERIFICATION.json` and `VERIFICATION_OPTIMIZED.json`; the unchanged inherited files are identified in `SOURCE_MANIFEST.json`.

The mathematical report is rebuilt by

```
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

A conventional LaTeX installation with the packages listed in the preamble is sufficient. Rebuilding the PDF can change its metadata and delivery hash without changing the mathematical source. The source and coefficient manifests remain the identity records for those components.

10.2 Suggested reading order

The central review question is whether each deletion or changed coordinate preserves the required integer domain in both directions. For P_{13}, first inspect recovery of the strong-root congruence, then the source bounds, then the exact indices, and finally the deleted quotients and their signs. In particular, the modulus $4qY/\gcd(4qY, H)$ is available before $q \mid X$; normalized strong divisibility is available before recovering the old main-root quotient. The constructive converse then checks every new positive coordinate.

For Q_{34}, inspect the restored source rows and the inverse root changes. The row $F_2 - f^2 = 0$ is intentionally retained, and the replacement auxiliary row differs by exactly $o^2(F_2 - f^2)$. The coefficient of $I^4 g^2$ gives a short exact degree certificate.

After these arguments, compare the literal JSON packets with the independent formulas and read the coefficient bindings and retained source proofs. The archive's root README and source manifest give the file inventory and provenance. This order keeps the mathematical equivalence, its finite representation, and the inherited compilation boundary separately reviewable.

References

[1] James P. Jones. *Universal Diophantine equation.* Journal of Symbolic Logic **47**(3) (1982), 549–571. doi:10.2307/2273588.

[2] Jonas Bayer, Marco David, Malte Hassler, Yuri Matiyasevich, and Dierk Schleicher. *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization.* arXiv:2506.20909v3, 22 October 2026. The version is pinned explicitly.

[3] Zhi-Wei Sun. *Ten unknowns for Hilbert's tenth problem over the integers.* arXiv:2609.01594v1, 1 September 2026. This is the specified first version, not an unversioned reference.

[4] ProveIt repository, source 34. *A mixed-radix, parity-separated, unrounded coefficient tuple.* Coefficient proof and complete bounded recipe. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[5] ProveIt repository, source 44. *Complement-packed native clauses: a smaller complete coefficient tuple.* Native compiler adapter and coefficient recipe. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[6] ProveIt repository, source 36. *Inherited Pell audit*. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[7] ProveIt repository, source 61. *Signed auxiliary-index recovery and canonical completeness*. The note retains its own earlier source pins; the inspected file is at repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[8] ProveIt repository, source 61. *Congruence carriers and a divided Pell norm*. Principal thirteen-positive degree-34 equation and its literal 93-gate circuit. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[9] ProveIt repository, source 62. *Sign recovery and centered norms for explicit ZFC consistency equations*. Thirty-five-integer sextic and thirty-integer octic comparison artifacts. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[10] ProveIt repository, source 64. *Structural reductions* and the forty-one-integer quartic proof. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[11] ProveIt repository, source 65. *Shared taps*, comparison results. Fourteen-positive degree-26, 92-operation endpoint. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[12] ProveIt repository, source 58. *Integral-root construction*, comparison results. Fifteen-positive degree-24, 90-operation endpoint. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[13] ProveIt repository, source 60. *Positive-gap construction*, transport companion. Fifteen-positive degree-24, 89-operation endpoint; its source-58 parent has 90 operations [12]. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[14] ProveIt repository. *Fixed blank source proof*. Native arithmetic and literal blank-machine finite-window semantics, with the transitive sources identified there. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[15] ProveIt repository. *Native arithmetic and the remaining calibration contract*. Identifies the external K215/refutation theorem and the remaining formal native-instance bridge. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.

[16] ProveIt repository. *The object-language ZFC obligation*. Exact distinction between the external equivalence and the internal `BProv` endpoint, including nonstandard-model obligations. Repository snapshot `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.